AI-Driven 6G Wireless Networks: A Comparative Taxonomy of Learning Paradigms, Applications, and Research Challenges

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Abstract*— The transition toward sixth-generation (6G) wireless networks is driven by ultra-low latency, massive connectivity, and heterogeneous services that conventional rule-based and optimization-centric management struggle to support. Artificial Intelligence (AI) has become a foundational element of proposed 6G architectures, ranging from deep learning and reinforcement learning to agentic and generative approaches based on large language models. This paper presents a literature-based comparative survey of AI techniques for 6G networks, covering network infrastructure, agentic AI-native architectures, physical-layer learning, network security, and application-specific optimization. A taxonomy of AI learning paradigms is constructed and mapped to major 6G applications, including radio access network intelligence, reconfigurable intelligent surfaces, integrated sensing and communication, non-terrestrial and UAV-assisted networks, and intrusion prediction for 6G-enabled Internet of Things (IoT). The reviewed works are compared in terms of data requirements, latency, scalability, and explainability. The comparison shows that generative and agentic AI are rapidly expanding but less standardized, while reinforcement learning and deep learning are comparatively mature. Key challenges include latency–reasoning trade-offs, energy-aware model placement, limited real-world training data, and the lack of unified security and governance frameworks. Research gaps and future directions are identified to support trustworthy, efficient, and standardized AI-driven 6G systems. This work provides a taxonomical synthesis of diverse AI-native approaches across the device-edge-core continuum.

Index Terms*— 6G, Artificial Intelligence, Agentic AI, Large Language Models, Reinforcement Learning, Federated Learning, Network Intelligence, Reconfigurable Intelligent Surfaces, Integrated Sensing and Communication, Wireless Networks.

I. INTRODUCTION

A. Background and Evolution of Wireless Networks

The evolution from 4G to 5G, 5G-Advanced, and now toward 6G has been driven by continually escalating requirements for data rate, latency, reliability, and connectivity density. Network complexity has grown accordingly, fuelled by the use of multiple frequency bands, massive device connectivity, non-terrestrial network (NTN) integration, small-cell densification, massive multiple-input multiple-output (MIMO) systems, and reconfigurable intelligent surfaces (RIS) [1]. Heterogeneous infrastructures, edge computing, network slicing, dynamic spectrum sharing, and increasingly stringent energy-efficiency and security requirements further compound this complexity, to the point where conventional operator-level and rule-based network management is no longer sufficient to manage the resulting system scale [1]. Optical fronthaul technologies and emerging physical-layer enablers such as free-space optics and light-fidelity (LiFi) are also being investigated to meet 6G's bandwidth demands, with AI increasingly proposed as a tool for managing the resulting fronthaul complexity [4].

B. Emergence of AI-Native 6G Networks

In response to this complexity, AI is increasingly framed not as an auxiliary optimization tool but as a foundational architectural element of 6G, a notion commonly referred to as AI-native networking [2], [12], [18]. Standardization bodies including 3GPP, IETF, ETSI, and the O-RAN Alliance have begun to converge on the idea that AI agents should be capable of intent understanding, contextual reasoning, and collaborative decision-making within the network itself, rather than functioning solely as isolated learning modules bolted onto individual network functions [2].

C. Motivation

Conventional optimization and rule-based approaches were adequate when network functions could be captured through explicit objective functions and bounded control loops. However, as 6G networks increasingly need to reconcile competing, ambiguous, or semantically expressed objectives, narrowly scoped optimization models become insufficient [2]. Likewise, testing and validating AI/ML-based network functions before deployment is itself a bottleneck: AI/ML model training in 6G depends on large volumes of real-world measurement data that are often unavailable prior to deployment, creating a circularity that traditional development pipelines struggle to resolve [1]. These observations motivate the shift toward more capable, proactive, and infrastructure-aware AI methods surveyed in this paper.

D. Research Gap

Several recurring limitations emerge across the reviewed literature. First, existing surveys and frameworks tend to address AI for 6G in a fragmented manner, focusing on isolated aspects such as orchestration, semantic communication, edge intelligence, or physical-layer adaptation, without providing a unified architecture or taxonomy [2], [12]. Related survey literature has itself been described as either too narrow in scope or outdated relative to recent breakthroughs in generative AI and semantic communication [18]. Second, agentic and LLM-based approaches, while rapidly emerging, lack standardized architectures, communication protocols, and governance mechanisms [2], [5]. Third, dataset and testbed limitations recur throughout the corpus: AI/ML training pipelines depend on scarce real-world telemetry [1], and even purpose-built cybersecurity frameworks have so far been validated on a single dataset [3]. Fourth, trust, security, and explainability remain underexplored relative to the pace of agentic AI adoption, with several works pointing to new attack surfaces that current telecom security frameworks do not yet cover [5], [16], [25].

E. Objectives of the Study

This survey aims to: 1) categorize the AI learning paradigms relevant to 6G represented in recent literature; 2) identify the algorithms, models, and architectures used by different researchers; 3) map AI techniques onto major 6G applications and enabling technologies; 4) compare approaches based on data requirements, latency, scalability, and explainability; 5) identify limitations and open research challenges reported across the literature; 6) synthesize recurring research gaps; and 7) highlight future research directions grounded in the reviewed studies.

F. Contributions

As a survey and comparative study, this paper contributes: a comparative taxonomy of AI learning paradigms represented in recent 6G literature; a literature-based mapping of AI techniques to 6G applications; a comparative analysis of reviewed approaches across multiple qualitative criteria; a synthesis of research challenges repeatedly identified across the corpus; and identification of research gaps and future directions. This paper does not claim to propose a new AI algorithm, protocol, or system; all technical contributions described herein originate from the cited literature.

G. Organization of the Paper

Section II reviews 6G requirements and enabling technologies. Section III presents the taxonomy of AI learning paradigms identified in the literature. Section IV maps AI techniques to 6G applications. Section V synthesizes a conceptual AI-enabled 6G architecture. Section VI provides a comparative analysis across paradigms. Section VII discusses research challenges. Section VIII presents research gaps and future directions. Section IX concludes the paper.

II. 6G WIRELESS NETWORKS: REQUIREMENTS AND ENABLING TECHNOLOGIES

A. 6G Vision and Requirements

The reviewed literature converges on a common set of 6G requirements: extremely high data rates, ultra-low latency, massive device connectivity, high reliability, energy efficiency, native intelligence, integrated sensing, and strengthened security [1], [18]. These requirements are not independent targets; several works emphasize that meeting them simultaneously is precisely what motivates the shift toward AI-native and, increasingly, agentic network intelligence [2].

B. 6G Network Architecture

Architecturally, the literature describes a continuing disaggregation trend: previously monolithic RAN functions are split into distinct components (Radio Unit, Distributed Unit, Central Unit under O-RAN), with a RAN Intelligent Controller applying AI/ML across different control time-scales [1]. Proposed AI-native 6G architectures layer semantic abstraction, agentic reasoning, and distributed multi-agent coordination above this deterministic infrastructure, separating reasoning from execution to preserve telecom-grade reliability guarantees [2].

C. Major Enabling Technologies

Reconfigurable Intelligent Surfaces (RIS): AI, particularly transformer-based deep learning, is used to predict received power and configure RIS elements in sub-6 GHz deployments in an energy-efficient manner [6].

Integrated Sensing and Communication (ISAC): AI enables sensing-as-a-service and joint sensing–communication resource optimization, including in energy-constrained IoT deployments [14], [17].

Non-Terrestrial Networks (NTN) and UAV-assisted communication: DRL and optimization-based beamforming support controller selection, connectivity, and RIS-assisted positioning; multi-modal foundation models unify learning across SAGIN domains [10], [15], [23], [24].

Open RAN (O-RAN) and testing infrastructure: Continuous-Testing (CT) pipelines generate real-world datasets required for AI/ML model training in O-RAN VNFs [1].

Massive/holographic MIMO and optical fronthaul: primarily hardware- and physical-layer-focused, with AI's role currently more peripheral (e.g., calibration, impairment compensation) [4], [11], [26].

D. Challenges of Conventional 6G Network Management

The literature consistently identifies why rule-based network management becomes untenable at 6G scale: network scale and heterogeneity, non-stationary channel conditions, large state/action spaces, strict real-time constraints, distributed device–edge–core infrastructure, and multi-objective trade-offs that cannot be reduced to a single optimization target [1], [2]. These challenges establish the motivation for the AI paradigms surveyed in Section III.

III. TAXONOMY OF AI LEARNING PARADIGMS FOR 6G

This section synthesizes the AI learning paradigms represented across the reviewed literature. Agentic AI and large language model (LLM)-based approaches are the most extensively represented paradigm in the collected corpus; reinforcement learning, deep learning, and federated/distributed learning are represented with moderate depth; paradigms such as pure unsupervised learning, semi-supervised learning, causal AI, and neurosymbolic AI are not substantively represented and are treated only briefly as emerging or largely absent directions, rather than given disproportionate coverage.

A. Supervised and Deep Learning

Deep learning appears primarily as the substrate for physical-layer prediction tasks. The Row-Vector Transformer (RVT) applies a transformer-based deep architecture to predict received power for sub-6 GHz RIS deployments in an energy-efficient manner, positioned as an alternative to conventional CNN/LSTM baselines [6]. Graph Neural Networks (GNNs) are proposed as a distinct deep learning family suited to topology-dependent wireless relationships, though standalone GNNs are noted as insufficient for continual learning and data scarcity, motivating hybrid designs (Section III-C) [12]. Purely unsupervised or semi-supervised learning is not substantively represented in the collected literature.

B. Reinforcement Learning and Deep Reinforcement Learning

RL is represented through concrete algorithmic instances. A lightweight Q-learning-based routing protocol is proposed for dynamic MANETs in 6G-oriented environments, targeting adaptive routing under topology change with low overhead [19]. DRL is applied to UAV positioning for RIS-assisted 6G emergency communications, with survivor-aware trajectory optimization learned via deep Q-learning in NOMA-enabled settings [24]. RL/DRL is favored for sequential decision problems with well-defined state–action–reward structure, in contrast with the more semantically-driven problems addressed by agentic AI (Section III-D).

C. Multi-Agent RL and Distributed/Federated Learning

MARL and federated/distributed learning are represented as complementary components within broader surveys. The D2D survey incorporates federated learning and MARL within its taxonomy for interference management, power control, and resource allocation [7]. The hybrid GNN-centric survey identifies federated learning as one of several paradigms — alongside DRL, meta-learning, generative models, and mixture-of-experts — combinable with GNNs to address continual learning and data scarcity [12]. Federated/distributed learning is motivated primarily by privacy preservation and reduced communication overhead in edge-heavy 6G-IoT deployments [17].

D. Generative AI, Large Language Models, and Agentic AI

Generative AI, LLMs, and agentic AI form the largest and most rapidly growing paradigm in the reviewed corpus. Agentic AI-native 6G is proposed as a four-layer architecture in which LLM-based agents act as bounded, policy-governed reasoning entities above deterministic 3GPP infrastructure, with empirical evaluation showing a trade-off between reasoning capability and system efficiency [2]. Small Language Models (SLMs) are applied to packet-level intrusion forecasting in a multi-agent cybersecurity framework, combined with a lightweight gradient-boosting detector and a knowledge-graph-driven mitigation agent [3]. A dedicated benchmark, 6G-Bench, evaluates semantic communication and network-level reasoning of foundation models in AI-native 6G [13]. A ReAct framework combining multimodal LLMs with retrieval-augmented generation is proposed for intent-driven 6G networking [9], while cognitive-bias considerations in agentic, LLM-driven autonomous 6G networks are examined separately [8]. Agentic AI is further explored for adaptive, dynamic security in programmable 6G networks [21] and for multimodal, trustworthy intelligence more broadly [25].

E. Explainable, Trustworthy, and Secure AI

Explainability, trust, and security form a distinct, increasingly urgent thread closely coupled to agentic AI. A dedicated survey catalogs emerging attack vectors, vulnerabilities, and defenses specific to agentic AI in 6G, examining both exploitation of autonomous agents and their own new attack surfaces [5]. A broader survey of adversarial threats in AI-native 6G catalogs adversarial machine learning risks and robustness considerations [16]. LLM-driven approaches are also proposed defensively for security and resilience in mission-critical 6G communications, including NTN and public-safety contexts [22]. XAI is referenced within broader AI-for-6G taxonomies [18] but does not yet appear as a standalone, deeply developed technique in this corpus; it is more often embedded within trust- and governance-oriented agentic AI discussions [5], [8], [25].

F. Causal and Neurosymbolic AI

Causal AI and neurosymbolic AI are not represented in the collected literature and are noted here only as plausible long-horizon research directions.

TABLE I. AI LEARNING PARADIGMS AND THEIR ROLE IN 6G

Paradigm	Role / 6G Application	Refs.
Deep Learning	RIS power prediction, channel modeling	[6],[12]
RL / DRL	MANET routing, UAV trajectory planning	[19],[24]
Federated / MARL	D2D & edge-IoT distributed coordination	[7],[12],[17]
Agentic AI / LLMs	Orchestration, IoT security, intent-driven comms	[2],[3],[5],[8],[9], [13],[21],[22],[25]
XAI / Trust	Security auditing, bias & mitigation mapping	[5],[8],[16],[25]

IV. AI APPLICATIONS IN 6G WIRELESS NETWORKS

A. Physical-Layer Communications and RIS

AI is applied at the physical layer principally for RIS-related channel and power prediction. The RVT model predicts received power for sub-6 GHz RIS deployments using a transformer-based architecture, framed around energy efficiency relative to conventional CNN/LSTM channel estimators [6].

B. Integrated Sensing and Communication (ISAC)

ISAC applications span network-for-sensing and sensing-as-a-service concepts [14], lightweight AI-driven adaptive resource allocation for energy-efficient ISAC in 6G-IoT [17], and active–passive RIS beamforming optimization for emergency UAV-ISAC [23].

C. Resource Allocation, D2D, and Edge Intelligence

The D2D survey maps deep learning, DRL, GNNs, federated learning, and MARL onto interference management, power control, and resource allocation for device-to-device communication in 5G/6G [7]. Edge intelligence is addressed via lightweight ISAC resource-allocation [17] and CT/DevOps infrastructure generating edge-relevant AI/ML datasets [1].

D. O-RAN, Network Automation, and Testing

A proposed 6G DevOps methodology establishes Continuous Testing (CT) as a critical enabler for VNF deployment validation and AI/ML dataset generation, demonstrated via a case study profiling an open-source CU+DU VNF [1]. Agentic AI is separately proposed for adaptable, dynamic security in programmable networks, extending automation into runtime security orchestration [21].

E. NTN and UAV-Assisted Networks

UAV-assisted vehicular networks are studied for AI-enabled controller selection and multi-band connectivity [15]; UAV-ISAC systems are optimized for emergency scenarios via active–passive beamforming [23]; DRL is applied to survivor-aware UAV positioning for RIS-assisted emergency communications [24]. Multi-modal foundation models unify learning across SAGIN domains [10].

F. Security, Privacy, and Trust

Security applications span knowledge-graph-driven IoT mitigation [3], surveys of attack/defense dynamics for agentic AI [5] and broader adversarial threats [16], zero-trust proximity-service credentials [20], agentic AI for adaptive security [21], LLM-driven security for mission-critical 6G [22], and secure multimodal agentic intelligence [25].

G. Semantic and Intent-Driven Communication

A ReAct framework combining multimodal LLMs with retrieval-augmented generation supports intent-driven 6G networking [9]; 6G-Bench evaluates semantic communication and network-level reasoning of foundation models in AI-native 6G [13].

TABLE II. AI TECHNIQUES MAPPED TO 6G APPLICATIONS

AI Technique	6G Application	Refs.
Transformer (RVT)	RIS received-power prediction	[6]
GNN / Hybrid GNN	Topology-aware network control	[12]
Q-learning / DQN	MANET routing, UAV positioning	[19],[24]
FL / MARL	D2D resource allocation, edge-IoT	[7],[12],[17]
LLM Agents / SLMs	Orchestration, IoT security, semantic comms	[2],[3],[9],[13]
Agentic Security / XAI	Autonomous defense, trust, bias auditing	[5],[8],[16],[21], [22],[25]
AI for ISAC	Sensing-as-a-service, UAV-ISAC	[14],[17],[23]

V. AI-ENABLED 6G NETWORK ARCHITECTURE

Synthesizing architectural elements described across the reviewed literature, a conceptual AI-enabled 6G architecture can be organized into five interacting layers. This synthesis is assembled from architectural elements described across the corpus [1], [2], [12], and is presented as a comparative organizing device rather than a novel proposal.

1) Application and Services Layer: intent-driven services, IoT security applications, semantic communication endpoints [3], [9], [13].
2) Agentic Reasoning and Orchestration Layer: LLM/SL based agents performing intent interpretation, planning, tool invocation, and reflection, operating above deterministic infra structure as policy-governed entities [2], [8], [21].
3) Distributed Multi-Agent / Edge Intelligence Layer: device edge-, and core-level agents coordinating via federated learn ing, MARL, and hybrid GNN-based control [7], [12], [17].
4) Deterministic 6G Infrastructure Layer: RAN, O-RAN, RIS, ISAC, NTN/UAV subsystems, and CT pipelines generating telemetry and enforcing real-time execution [1], [6], [14], [23].
5) Trust, Governance, and Security Layer: cross-cutting autho rization, adversarial-robustness validation, and mitigation rec ommendation spanning all layers above [5], [16], [20], [25].

A consistent theme is the separation of reasoning from execu tion: agentic components propose actions, while a distinct governance/enforcement layer validates and authorizes them before they reach deterministic network-control interfaces [2], [3].

VI. COMPARATIVE ANALYSIS OF AI FOR 6G

Direct quantitative comparison across the reviewed papers is not possible, since they employ different datasets, simulation environments, and metrics; this limitation is stated explicitly rather than resolved through fabricated benchmarking. This section instead offers a qualitative, literature-grounded comparison.

Regarding data requirements, deep learning and transformer-based physical-layer models typically require substantial labeled or simulated channel data [6], whereas RL depends on interaction with a simulated or real environment [19], [24]. Federated/distributed learning trades off performance and communication efficiency for data locality and privacy [7], [12]. Agentic and LLM-based approaches depend heavily on pretraining and, where fine-tuned, on benchmark or synthetic scenario data [3], [13].

Regarding latency, RL/DRL and lightweight deep learning are generally reported as compatible with real-time control loops [6], [19], [24]. Agentic, LLM-based reasoning introduces a latency–reasoning trade-off: no single language-model configuration simultaneously satisfies latency, throughput, and accuracy, motivating heterogeneous, layered deployment [2].

Regarding scalability, GNN-hybrid and federated/MARL approaches are motivated by distributed, large-scale deployment [7], [12], while agentic multi-agent fabrics raise open coordination questions, since no unified agent-to-agent communication protocol yet exists [2], [5].

Regarding explainability, security, and trust, this remains the least mature dimension across all paradigms. Agentic AI introduces new, only partially understood risks, including inherited cognitive biases [8], hallucination-driven instability, and expanded attack surfaces [5], [16], [25].

TABLE III. COMPARATIVE ANALYSIS OF AI PARADIGMS FOR 6G

Paradigm	Data Req.	Latency Fit	Maturity
Deep Learning	Moderate–High	High	Established
RL / DRL	Interaction-driven	High	Established
Federated / MARL	Distributed	Moderate	Emerging
Agentic AI / LLMs	Pretrain+fine-tune	Low–Mod.	Least mature
XAI / Trust	Policy/KG-based	Moderate	Least mature

VII. RESEARCH CHALLENGES AND OPEN ISSUES

A. Data Availability and Dataset Quality

AI/ML model training for 6G VNFs depends on large-scale real-world measurement data frequently unavailable prior to deployment, motivating CT pipelines [1]. Purpose-built security frameworks report reliance on single datasets, raising generalization questions [3].

B. Computational Complexity and Latency

Agentic, LLM-based reasoning introduces latency costs scaling with model size and reasoning depth, tensioning against sub-millisecond 6G control timing [2].

C. Energy Efficiency

Energy efficiency recurs from RIS power-prediction [6], to lightweight ISAC-IoT resource allocation [17], to concerns about large-scale agentic reasoning cost [2].

D. Non-Stationary and Dynamic Environments

D2D and mobile environments change continuously, requiring RL-based routing and resource allocation to adapt online [7], [19].

E. Privacy and Distributed Learning

Federated/distributed learning is motivated by privacy in edge-heavy deployments, but communication overhead and non-IID data remain constraints [7], [12].

F. Security and Adversarial Robustness

Agentic AI broadens the 6G attack surface (prompt injection, telemetry poisoning, agent impersonation) beyond traditional protocol-layer security [5], [16]. Zero-trust mechanisms are explored for proximity services [20].

G. Explainability, Trust, and Cognitive Bias

Cognitive-bias analysis of agentic 6G systems shows reasoning agents can behave unpredictably in ways not captured by conventional accuracy metrics [8].

H. Generative AI Reliability

Hallucination and inconsistent reasoning are open risks for LLM-based network agents influencing real network configuration [2], [8].

I. Scalability of Multi-Agent Systems

Coordinating distributed multi-agent fabrics raises unresolved questions about discovery, identity, and communication standardization [2], [5].

J. Standardization and Interoperability

No unified agentic AI-native architecture has been adopted by 3GPP, ITU-R, ETSI, IETF, or O-RAN despite converging interest [2], [5].

K. AI–Hardware Co-Design

RF transceiver non-idealities, holographic MIMO, and optical fronthaul indicate AI's physical-layer impairment-compensation role remains underdeveloped [4], [11], [26].

VIII. RESEARCH GAPS AND FUTURE DIRECTIONS

A. Identified Research Gaps

1) Lack of a unified AI taxonomy and architecture for 6G [2], [12], [18].

2) Absence of standardized agentic architectures and agent-to-agent protocols [2], [5].

3) Dataset and testbed limitations constraining AI/ML training and validation [1], [3], [13].

4) Underdeveloped trust, security, and governance mechanisms for agentic AI [5], [8], [16], [25].

5) Limited AI–hardware co-design for physical-layer impairment compensation [26].

B. Future Research Directions

- Standardized, interoperable agentic architectures and agent-to-agent protocols aligned with 3GPP/IETF/ETSI/O-RAN efforts [2], [5].
- Heterogeneous, latency-aware deployment placing lighter models at device/edge and capable reasoning agents at the core [2].
- Expanded 6G AI benchmarks and CT pipelines to reduce dependence on scarce real-world data [1], [3], [13].
- Integration of explainability and cognitive-bias auditing directly into agentic reasoning pipelines [8].
- Extension of hybrid GNN-centric and federated/MARL architectures across the device–edge–core continuum [7], [12].

- Deeper AI–hardware co-design for RF impairment compensation, RIS control, and holographic MIMO [6], [11], [26].

IX. CONCLUSION

This survey has examined recent literature on AI for 6G wireless networks through a comparative taxonomy of learning paradigms, an application-level mapping, a synthesized architectural view, and a qualitative comparative analysis. The reviewed corpus reveals a field concentrated heavily around agentic AI and large language models, applied to network orchestration, intent-driven communication, and security, alongside a more established but narrower base of reinforcement learning, deep learning, and federated/distributed learning applied to resource allocation, physical-layer prediction, and distributed coordination. Recurring challenges include a latency–reasoning trade-off unique to agentic systems, persistent data and testbed scarcity, and an explainability and security gap that has not kept pace with the rapid adoption of generative and agentic AI. These findings point toward standardized agentic architectures, heterogeneous latency-aware deployment, expanded benchmarking, embedded explainability, and deeper AI–hardware co-design as concrete future directions for AI-native 6G networks.

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